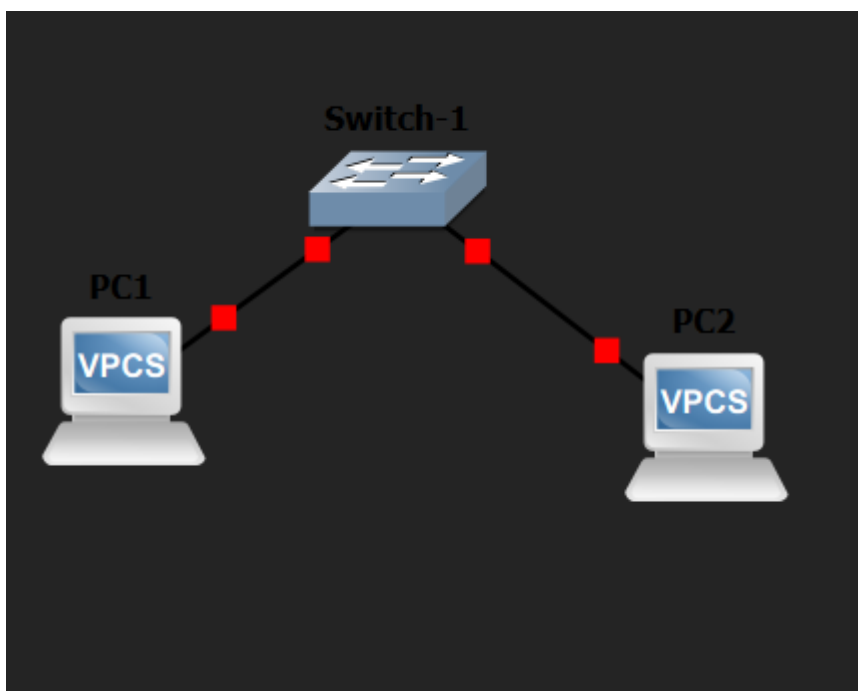
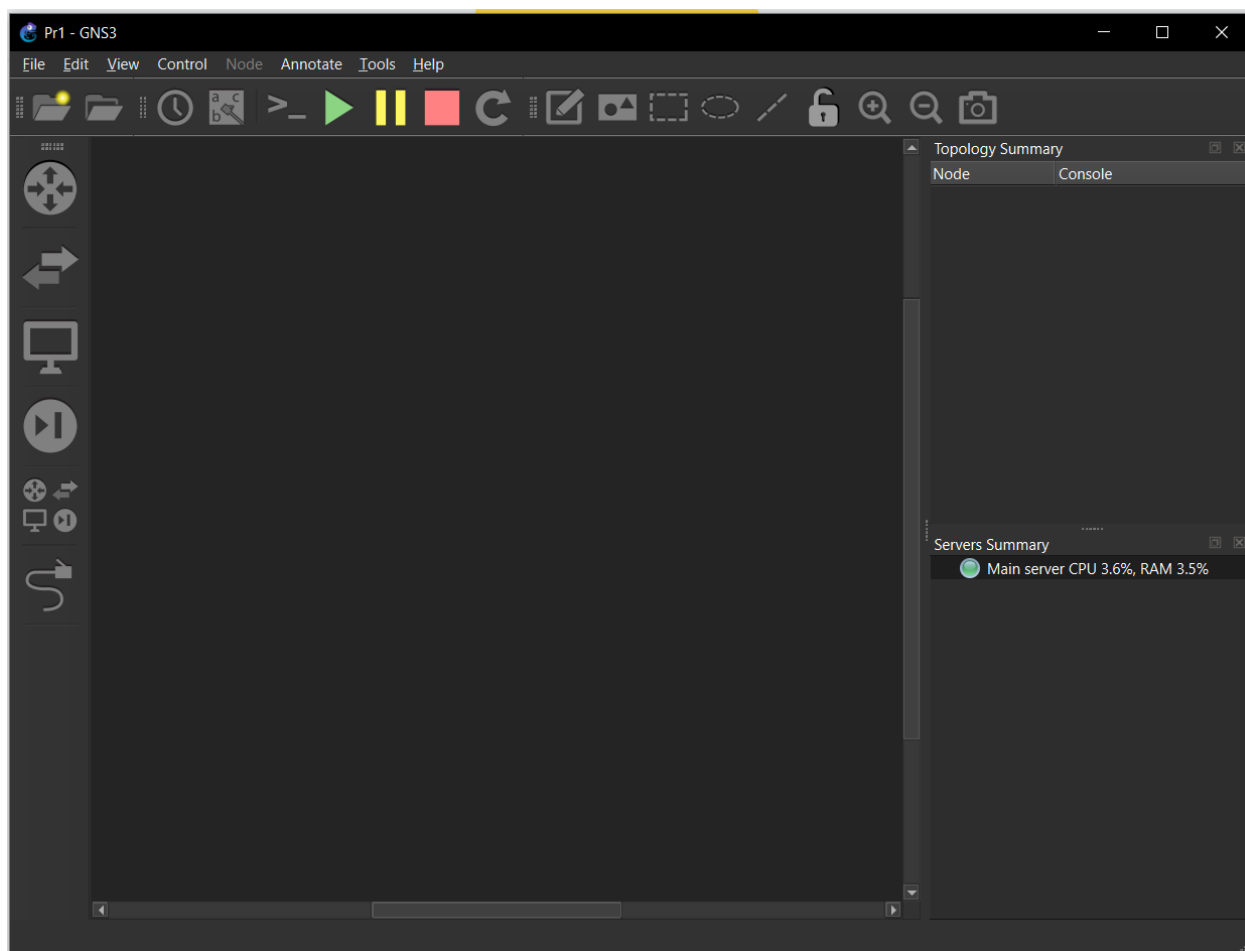


Модуль 4. Практика 1

Абламский Н.Е.



```
PC1> ip 192.168.10.10/24 192.168.10.200
Checking for duplicate address...
PC1 : 192.168.10.10 255.255.255.0 gateway 192.168.10.200
```

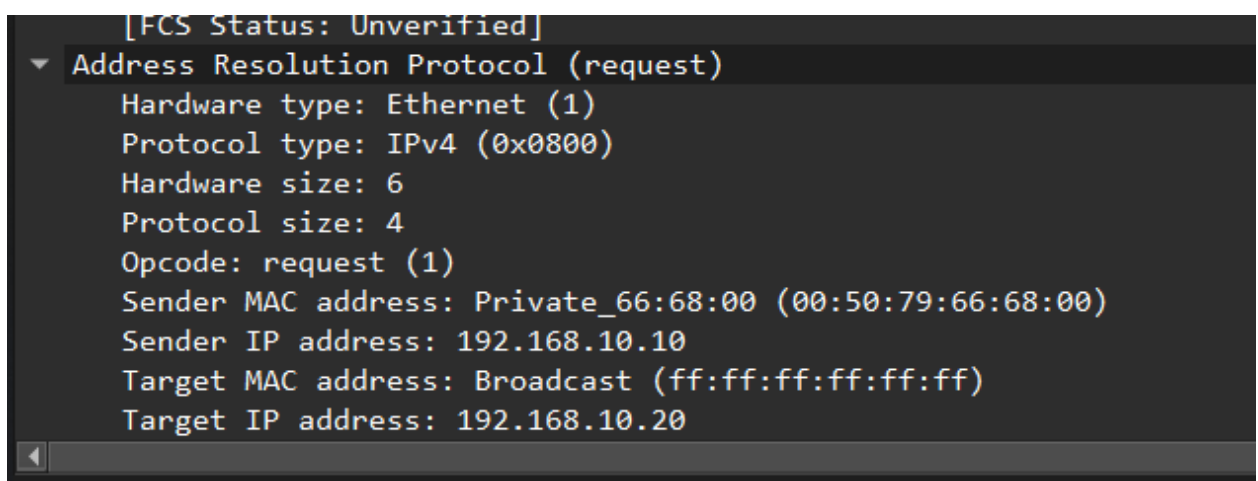
```
PC2> ip 192.168.10.20/24 192.168.10.200
Checking for duplicate address...
PC2 : 192.168.10.20 255.255.255.0 gateway 192.168.10.200
```

```
PC1> ping 192.168.10.20
```

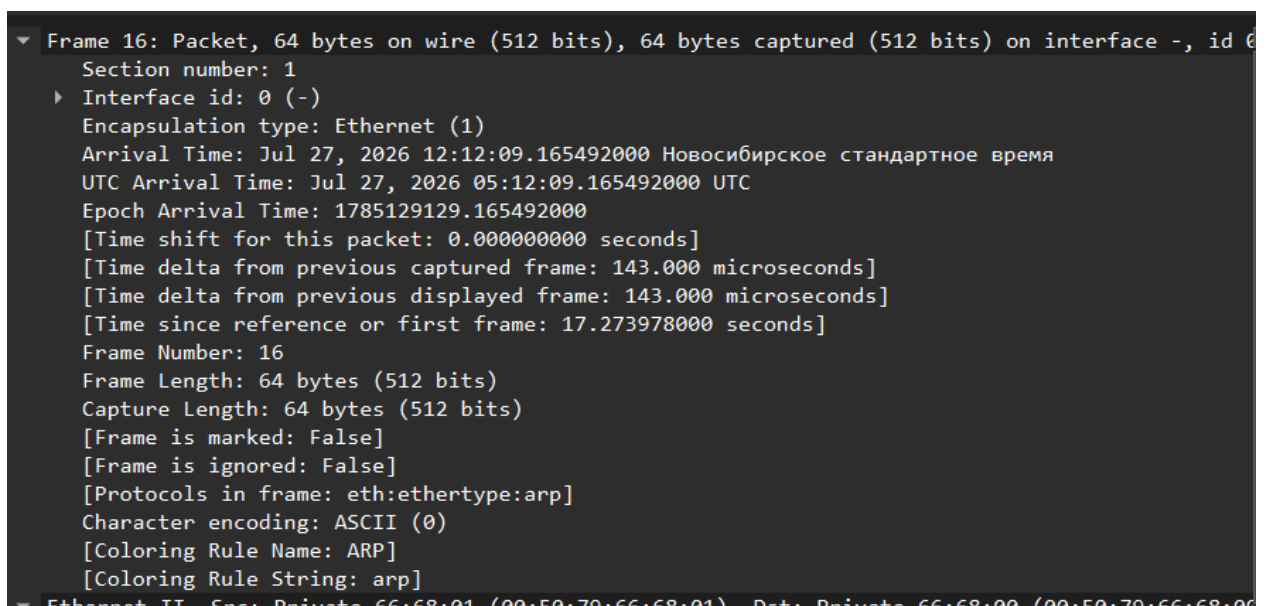
```
84 bytes from 192.168.10.20 icmp_seq=1 ttl=64 time=0.437 ms
84 bytes from 192.168.10.20 icmp_seq=2 ttl=64 time=0.615 ms
84 bytes from 192.168.10.20 icmp_seq=3 ttl=64 time=0.655 ms
84 bytes from 192.168.10.20 icmp_seq=4 ttl=64 time=3.563 ms
```

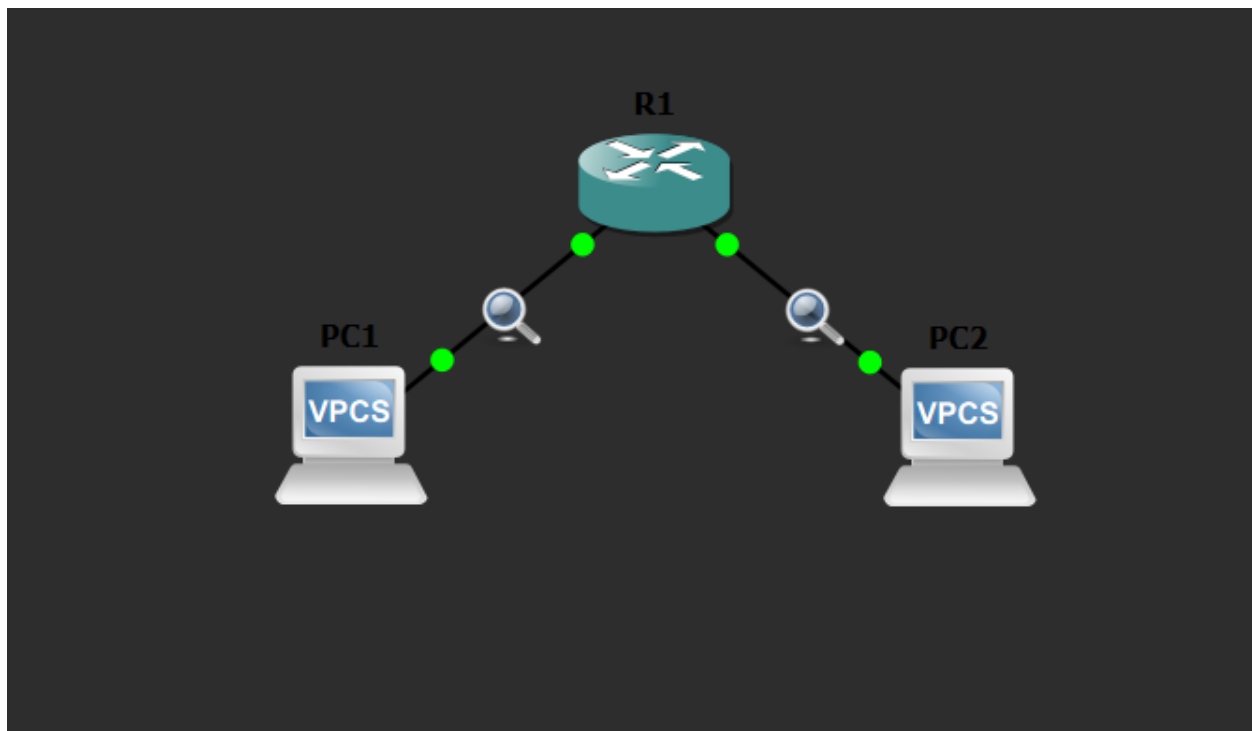
Запрос:

No.	Time	Source	Destination	Protocol	Length	Info
15	17.273835	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.20? Tell 192.168.10.10
16	17.273978	Private_66:68:01	Private_66:68:00	ARP	64	192.168.10.20 is at 00:50:79:66:68:01



ПК1 спрашивает у всех в сети (сообщение barodcast) у кого IPv4=192.168.10.20 (протокол arp), и просит прислать ответ отправителю (с IP=192.168.10.10). Ответ приходит после:





No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.20? (ARP Probe)
2	1.000654	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.20? (ARP Probe)
3	2.001309	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.20? (ARP Probe)
4	135.195239	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.1.1.70 (Request)

```
#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status
Protocol				
FastEthernet0/0	unassigned	YES	unset	administratively
down down				
FastEthernet1/0	unassigned	YES	unset	administratively
down down				
Ethernet2/0	unassigned	YES	unset	administratively
down down				
Ethernet2/1	unassigned	YES	unset	administratively
down down				
Ethernet2/2	unassigned	YES	unset	administratively
down down				
Ethernet2/3	unassigned	YES	unset	administratively
down down				
Serial3/0	unassigned	YES	unset	administratively
down down				
Serial3/1	unassigned	YES	unset	administratively
down down				
Serial3/2	unassigned	YES	unset	administratively
down down				
Serial3/3	unassigned	YES	unset	administratively
down down				

```
R1(config)#interface range ethernet2/0 - 1
R1(config-if-range)#no sh
R1(config-if-range)#no shutdown
```

```
PC1> ip 100.170.10.20/24 100.170.10.200
Checking for duplicate address...
PC1 : 100.170.10.20 255.255.255.0 gateway 100.170.10.200
```

```
PC2> ip 192.168.10.10/24 192.168.10.200
Checking for duplicate address...
PC2 : 192.168.10.10 255.255.255.0 gateway 192.168.10.200
```

```
R1(config)#interface ethernet2/0
R1(config-if)#ip address 100.170.10.200 255.255.255.0
```

```
R1(config)#interface ethernet2/1
R1(config-if)#ip address 192.168.10.200 255.255.255.0
```

```
R1#copy run startup-config
Destination filename [startup-config]?
Building configuration...
```

```
ping 192.168.10.10
```

```
192.168.10.10 icmp_seq=1 timeout
84 bytes from 192.168.10.10 icmp_seq=2 ttl=63 time=11.032 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=63 time=15.277 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=63 time=15.346 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=63 time=15.587 ms
```

ЛИНК ПК1 – R1

No.	Time	Source	Destination	Protocol	Length	Info
2	0.988058	Private 66:68:01	Broadcast	ARP	64	who has 100.170.10.20? Tell 100.170.10.20
3	0.994046	cc:01:06:4b:00:20	Private 66:68:01	ARP	60	100.170.10.200 is at cc:01:06:4b:00:20
4	0.994399	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1e9d, seq=1/256, tt
5	1.024845	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1e9d, seq=1/256, tt
6	2.025999	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1f9d, seq=2/512, tt
7	2.041211	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1f9d, seq=2/512, tt
9	3.042266	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x219d, seq=3/768, tt
10	3.061408	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x219d, seq=3/768, tt
11	4.062780	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x229d, seq=4/1024, t
12	4.084157	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x229d, seq=4/1024, t
13	5.085690	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x239d, seq=5/1280, t
14	5.096898	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x239d, seq=5/1280, t

```

Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
  Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
    Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    Source: Private_66:68:01 (00:50:79:66:68:01)
      Type: ARP (0x0806)
      [Stream index: 1]
      Padding: 0000000000000000000000000000000000000000
      Frame check sequence: 0x00000000 [unverified]
      [FCS Status: Unverified]
  Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
    Sender IP address: 100.170.10.20
    Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
    Target IP address: 100.170.10.200

```

No.	Time	Source	Destination	Protocol	Length	Info
2	0.988058	Private 66:68:01	Broadcast	ARP	64	Who has 100.170.10.200? Tell 100.170.10.20
3	0.994046	cc:01:06:4b:00:20	Private 66:68:01	ARP	60	100.170.10.200 is at cc:01:06:4b:00:20
4	0.994399	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1e9d, seq=1/256, ttl=64 (reply in 5)
5	1.024845	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1e9d, seq=1/256, ttl=63 (request in 4)
6	0.205999	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1f9d, seq=2/512, ttl=64 (reply in 7)
7	2.041211	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1f9d, seq=2/512, ttl=63 (request in 6)
9	3.042266	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x219d, seq=3/768, ttl=64 (reply in 10)
10	3.061408	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x219d, seq=3/768, ttl=63 (request in 9)
11	4.062780	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x229d, seq=4/1024, ttl=64 (reply in 12)
12	4.084157	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x229d, seq=4/1024, ttl=63 (request in 11)
13	5.085690	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x239d, seq=5/1280, ttl=64 (reply in 14)
14	5.096898	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x239d, seq=5/1280, ttl=63 (request in 13)

```

▶ Frame 3: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
  ▶ Ethernet II, Src: cc:01:06:4b:00:20 (cc:01:06:4b:00:20), Dst: Private_66:68:01 (00:50:79:66:68:01)
    ▶ Destination: Private_66:68:01 (00:50:79:66:68:01)
      ▶ Source: cc:01:06:4b:00:20 (cc:01:06:4b:00:20)
        Type: ARP (0x0806)
        [Stream index: 2]
        Padding: 00000000000000000000000000000000
      ▶ Address Resolution Protocol (reply)
        Hardware type: Ethernet (1)
        Protocol type: IPv4 (0x0800)
        Hardware size: 6
        Protocol size: 4
        Opcode: reply (2)
        Sender MAC address: cc:01:06:4b:00:20 (cc:01:06:4b:00:20)
        Sender IP address: 100.170.10.200
        Target MAC address: Private_66:68:01 (00:50:79:66:68:01)
        Target IP address: 100.170.10.20

```

No.	Time	Source	Destination	Protocol	Length	Info
2	0.988058	Private_66:68:01	Broadcast	ARP	64	Who has 100.170.10.200? Tell 100.170.10.20
3	0.994046	cc:01:06:4b:00:20	Private_66:68:01	ARP	60	100.170.10.200 is at cc:01:06:4b:00:20
→ 4	0.994399	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1e9d, seq=1/256, ttl=64 (reply in 5)
← 5	1.024845	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1e9d, seq=1/256, ttl=63 (request in 4)
6	2.025999	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1f9d, seq=2/512, ttl=64 (reply in 7)
7	2.041211	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1f9d, seq=2/512, ttl=63 (request in 6)
9	3.042266	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x219d, seq=3/768, ttl=64 (reply in 10)
10	3.061408	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x219d, seq=3/768, ttl=63 (request in 9)
11	4.062780	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x229d, seq=4/1024, ttl=64 (reply in 12)
12	4.084157	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x229d, seq=4/1024, ttl=63 (request in 11)
13	5.085690	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x239d, seq=5/1280, ttl=64 (reply in 14)
14	5.096898	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x239d, seq=5/1280, ttl=63 (request in 13)

▶ Frame 4: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
 ▾ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: cc:01:06:4b:00:20 (cc:01:06:4b:00:20)
 ▶ Destination: cc:01:06:4b:00:20 (cc:01:06:4b:00:20)
 ▶ Source: Private_66:68:01 (00:50:79:66:68:01)
 Type: IPv4 (0x0800)
 [Stream index: 2]
 ▶ Internet Protocol Version 4, Src: 100.170.10.20, Dst: 192.168.10.1
 ▾ Internet Control Message Protocol
 Type: Echo (ping) request (8)
 Code: 0
 Checksum: 0x016e [correct]
 [Checksum Status: Good]
 Identifier (BE): 7837 (0x1e9d)
 Identifier (LE): 40222 (0x9d1e)
 Sequence Number (BE): 1 (0x0001)
 Sequence Number (LE): 256 (0x0100)
 [Response frame: 5]
 ▾ Data (56 bytes)
 Data: 08090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
 [Length: 56]

No.	Time	Source	Destination	Protocol	Length	Info
2	0.988058	Private_66:68:01	Broadcast	ARP	64	Who has 100.170.10.200? Tell 100.170.10.20
3	0.994046	cc:01:06:4b:00:20	Private_66:68:01	ARP	60	100.170.10.200 is at cc:01:06:4b:00:20
→ 4	0.994399	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1e9d, seq=1/256, ttl=64 (reply in 5)
← 5	1.024845	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1e9d, seq=1/256, ttl=63 (request in 4)
6	2.025999	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1f9d, seq=2/512, ttl=64 (reply in 7)
7	2.041211	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1f9d, seq=2/512, ttl=63 (request in 6)
9	3.042266	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x219d, seq=3/768, ttl=64 (reply in 10)
10	3.061408	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x219d, seq=3/768, ttl=63 (request in 9)
11	4.062780	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x229d, seq=4/1024, ttl=64 (reply in 12)
12	4.084157	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x229d, seq=4/1024, ttl=63 (request in 11)
13	5.085690	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x239d, seq=5/1280, ttl=64 (reply in 14)
14	5.096898	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x239d, seq=5/1280, ttl=63 (request in 13)

▶ Frame 5: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
 ▾ Ethernet II, Src: cc:01:06:4b:00:20 (cc:01:06:4b:00:20), Dst: Private_66:68:01 (00:50:79:66:68:01)
 ▶ Destination: Private_66:68:01 (00:50:79:66:68:01)
 ▶ Source: cc:01:06:4b:00:20 (cc:01:06:4b:00:20)
 Type: IPv4 (0x0800)
 [Stream index: 2]
 ▶ Internet Protocol Version 4, Src: 192.168.10.1, Dst: 100.170.10.20
 ▾ Internet Control Message Protocol
 Type: Echo (ping) reply (0)
 Code: 0
 Checksum: 0x096e [correct]
 [Checksum Status: Good]
 Identifier (BE): 7837 (0x1e9d)
 Identifier (LE): 40222 (0x9d1e)
 Sequence Number (BE): 1 (0x0001)
 Sequence Number (LE): 256 (0x0100)
 [Request frame: 4]
 [Response time: 30,446 ms]
 ▾ Data (56 bytes)
 Data: 08090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
 [Length: 56]

Линк ПК2 – R1

No.	Time	Source	Destination	Protocol	Length	Info
2	1.004199	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1e9d, seq=1/256, ttl=
3	1.004357	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.200? Tell 192.168.10.1
4	1.015537	cc:01:06:4b:00:21	Private_66:68:00	ARP	60	192.168.10.200 is at cc:01:06:4b:00:21
5	1.016319	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1e9d, seq=1/256, ttl=
6	2.031199	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1f9d, seq=2/512, ttl=
7	2.031363	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1f9d, seq=2/512, ttl=
9	3.051416	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x219d, seq=3/768, ttl=
10	3.052745	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x219d, seq=3/768, ttl=
11	4.073723	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x229d, seq=4/1024, ttl=
12	4.073888	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x229d, seq=4/1024, ttl=
13	5.086860	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x239d, seq=5/1280, ttl=
14	5.086958	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x239d, seq=5/1280, ttl=

▶ Frame 3: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
 ▶ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 ▶ Address Resolution Protocol (request)
 Hardware type: Ethernet (1)
 Protocol type: IPv4 (0x0800)
 Hardware size: 6
 Protocol size: 4
 Opcode: request (1)
 Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
 Sender IP address: 192.168.10.1
 Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
 Target IP address: 192.168.10.200

No.	Time	Source	Destination	Protocol	Length	Info
2	1.004199	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1e9d, seq=1/256, ttl=
3	1.004357	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.200? Tell 192.168.10.1
4	1.015537	cc:01:06:4b:00:21	Private_66:68:00	ARP	60	192.168.10.200 is at cc:01:06:4b:00:21
5	1.016319	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1e9d, seq=1/256, ttl=
6	2.031199	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1f9d, seq=2/512, ttl=
7	2.031363	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1f9d, seq=2/512, ttl=
9	3.051416	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x219d, seq=3/768, ttl=
10	3.052745	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x219d, seq=3/768, ttl=
11	4.073723	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x229d, seq=4/1024, ttl=
12	4.073888	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x229d, seq=4/1024, ttl=
13	5.086860	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x239d, seq=5/1280, ttl=
14	5.086958	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x239d, seq=5/1280, ttl=

▶ Frame 4: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
 ▶ Ethernet II, Src: cc:01:06:4b:00:21 (cc:01:06:4b:00:21), Dst: Private_66:68:00 (00:50:79:66:68:00)
 ▶ Address Resolution Protocol (reply)
 Hardware type: Ethernet (1)
 Protocol type: IPv4 (0x0800)
 Hardware size: 6
 Protocol size: 4
 Opcode: reply (2)
 Sender MAC address: cc:01:06:4b:00:21 (cc:01:06:4b:00:21)
 Sender IP address: 192.168.10.200
 Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
 Target IP address: 192.168.10.1

arp icmp						
No.	Time	Source	Destination	Protocol	Length	Info
2	1.004199	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1e9d, seq=1/256, ttl=63 (reply in 5)
3	1.004357	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.200? Tell 192.168.10.1
4	1.015537	cc:01:06:4b:00:21	Private_66:68:00	ARP	60	192.168.10.200 is at cc:01:06:4b:00:21
5	1.016319	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1e9d, seq=1/256, ttl=64 (request in 2)
→ 6	2.031199	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1f9d, seq=2/512, ttl=63 (reply in 7)
← 7	2.031363	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1f9d, seq=2/512, ttl=64 (request in 6)
9	3.051416	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x219d, seq=3/768, ttl=63 (reply in 10)
10	3.052745	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x219d, seq=3/768, ttl=64 (request in 9)
11	4.073723	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x229d, seq=4/1024, ttl=63 (reply in 12)
12	4.073888	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x229d, seq=4/1024, ttl=64 (request in 11)
13	5.086860	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x239d, seq=5/1280, ttl=63 (reply in 14)
14	5.086958	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x239d, seq=5/1280, ttl=64 (request in 13)

- Frame 6: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0 - Ethernet II, Src: cc:01:06:4b:00:21 (cc:01:06:4b:00:21), Dst: Private_66:68:00 (00:50:79:66:68:00) - Internet Protocol Version 4, Src: 100.170.10.20, Dst: 192.168.10.1 - Internet Control Message Protocol - Type: Echo (ping) request (8) - Code: 0 - Checksum: 0x006d [correct] [Checksum Status: Good] - Identifier (BE): 8093 (0x1f9d) - Identifier (LE): 40223 (0x9d1f) - Sequence Number (BE): 2 (0x0002) - Sequence Number (LE): 512 (0x0200) [Response frame: 7] - Data (56 bytes)	<pre> 0000 00 50 79 66 68 00 0010 00 54 9d 1f 9d 1f 0020 0a 01 08 00 00 00 0030 0e 0f 10 11 12 13 0040 1e 1f 20 21 22 23 0050 2e 2f 30 31 32 33 0060 3e 3f </pre>
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arp icmp						
No.	Time	Source	Destination	Protocol	Length	Info
2	1.004199	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1e9d, seq=1/256, ttl=63 (reply in 5)
3	1.004357	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.200? Tell 192.168.10.1
4	1.015537	cc:01:06:4b:00:21	Private_66:68:00	ARP	60	192.168.10.200 is at cc:01:06:4b:00:21
5	1.016319	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1e9d, seq=1/256, ttl=64 (request in 2)
→ 6	2.031199	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x1f9d, seq=2/512, ttl=63 (reply in 7)
← 7	2.031363	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x1f9d, seq=2/512, ttl=64 (request in 6)
9	3.051416	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x219d, seq=3/768, ttl=63 (reply in 10)
10	3.052745	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x219d, seq=3/768, ttl=64 (request in 9)
11	4.073723	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x229d, seq=4/1024, ttl=63 (reply in 12)
12	4.073888	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x229d, seq=4/1024, ttl=64 (request in 11)
13	5.086860	100.170.10.20	192.168.10.1	ICMP	98	Echo (ping) request id=0x239d, seq=5/1280, ttl=63 (reply in 14)
14	5.086958	192.168.10.1	100.170.10.20	ICMP	98	Echo (ping) reply id=0x239d, seq=5/1280, ttl=64 (request in 13)

- Frame 7: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0 - Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:06:4b:00:21 (cc:01:06:4b:00:21) - Internet Protocol Version 4, Src: 192.168.10.1, Dst: 100.170.10.20 - Internet Control Message Protocol - Type: Echo (ping) reply (0) - Code: 0 - Checksum: 0x086d [correct] [Checksum Status: Good] - Identifier (BE): 8093 (0x1f9d) - Identifier (LE): 40223 (0x9d1f) - Sequence Number (BE): 2 (0x0002) - Sequence Number (LE): 512 (0x0200) [Request frame: 6] [Response time: 0,164 ms] - Data (56 bytes)	
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Когда ПК1 из сети 100.170.10.0/24 пытается пинговать ПК2 из сети 192.168.10.0/24, он понимает, что получатель находится в другой подсети, поэтому пакет надо отправлять на шлюз по умолчанию. Для этого ПК1 сначала выполняет ARP-запрос для IP-адреса шлюза (100.170.10.200), чтобы узнать MAC-адрес интерфейса маршрутизатора. После получения ответа ПК1 отправляет ICMP-запрос на MAC-адрес маршрутизатора. Маршрутизатор, получив пакет, смотрит в свою таблицу маршрутизации и определяет, что для сети 192.168.10.0/24 пакет нужно отправить через другой интерфейс. На этом интерфейсе также выполняется ARP-запрос для IP-адреса ПК2, затем ICMP-запрос доставляется ПК2, а ответ идёт в обратном порядке. Важно отметить, что при отправке ICMP-ответа ПК2 также сталкивается с тем, что целевой IP (ПК1) находится в другой сети, поэтому он не может отправить кадр напрямую. Вместо этого ПК2 выполняет ARP-запрос для своего шлюза по умолчанию (192.168.10.200), чтобы узнать его MAC-адрес. Именно этот ARP-запрос с Target IP 192.168.10.200 перехватывается на втором линке (между ПК2 и роутером) и виден на скриншоте. Таким образом, на каждом линке ARP-кадры всегда направлены на поиск локального шлюза, а не конечного получателя. На каждом линке можно перехватить соответствующие ARP-кадры и ICMP-пакеты, что позволяет проследить полный путь данных.